CANADIAN PATENT APPLICATION

Canadian Intellectual Property Office (CIPO)

Patent Act, R.S.C. 1985, c. P-4

SPECIFICATION

Title:	System and Method for Opportunistic Multimodal Constraint-Based Input Disambiguation Using Non-Recording Wearable Devices
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Country of Residence:	Canada
Filed:	2026
Status:	Draft Application (unpublished)
Classification:	CPC G06F 3/01; G06F 3/041; G06F 3/16

Abstract

A system and method are disclosed for improving the accuracy and robustness of touchscreen and gesture-based input by selectively incorporating auxiliary, non-recording wearable sensors that supply real-time constraint data for disambiguating user intent. Under normal operating conditions the system relies exclusively on the primary input surface and introduces no additional overhead or dependency. When primary-signal confidence falls below a dynamically computed threshold—due to environmental factors such as precipitation or vibration, or to weak-signal conditions such as gloved operation—the system activates a secondary wearable constraint device (exemplified herein as lightweight glasses equipped with optical or infrared sensing) and a tertiary haptic-glove input channel. The wearable device does not store or transmit persistent visual data and performs only transient, real-time constraint evaluation. Input events are accepted or rejected based on cross-modal agreement rather than single-surface inference. The system further supports screen-free haptic input when postural context, as assessed by the wearable device, indicates a favourable configuration. The architecture degrades gracefully when auxiliary devices are unavailable, absent, or inactive.

Field of the Invention

The present invention relates to human-computer interaction systems, and more particularly to multimodal input systems that integrate wearable sensing devices with touch interfaces to improve input accuracy, reduce false positives, and enable additional interaction modalities, while preserving privacy through the principled exclusion of recording, storage, and display capabilities from auxiliary devices.

Background of the Invention

Limitations of Capacitive Touchscreen Inference

Capacitive touchscreens infer user intent from perturbations in an electrostatic field. While effective under controlled conditions, such systems exhibit degraded performance whenever non-intentional stimuli produce perturbations of similar magnitude or spatial distribution to finger contact. Representative failure cases include rain, condensation, conductive contaminants, unintended palm contact, and operation with gloved hands or non-conductive implements.

Existing mitigations—spatial filtering, temporal thresholding, specialised oleophobic or moisture-repelling coatings, and software-defined "glove modes"— remain confined to the local surface measurement. Because they have access to only one sensing modality, such systems cannot fundamentally discriminate between intentional and non-intentional stimuli when both produce similar signatures at the surface grid. The problem is not one of signal fidelity; it is one of underdetermination. A raindrop impacting a glass surface and a fingertip approaching deliberately are, from the perspective of the electrostatic grid alone, partially indistinguishable.

Inadequacy of Single-Modality Confidence Estimation

Current devices attempt to estimate touch confidence from signal strength, contact area, and temporal profile. These heuristics are adequate in calm environments but degrade non-gracefully under compound adverse conditions. A device subjected simultaneously to rain and vibration (as on a train or during turbulent flight) may generate a continuous stream of spurious events that overwhelm any single-surface filter.

Prior Art in Gesture Recognition and Smart Glasses

Optical and depth-based gesture recognition systems have been explored for gaming, augmented reality, and touchless interface applications. However, such systems are typically designed as primary

input modalities, replacing touch rather than augmenting it, and depend on continuous scene capture, stored video streams, and complex real-time scene reconstruction. Their reliance on persistent recording raises significant privacy concerns.

Augmented reality glasses and smart glasses similarly include cameras, persistent recording capabilities, display surfaces, and often full scene-understanding pipelines. These architectural choices create social friction—bystanders cannot determine whether they are being recorded—and introduce technical complexity that far exceeds the requirements of reliable touch disambiguation.

Haptic Glove Input Systems

Air-typing and haptic-glove systems have been explored as alternatives to physical keyboards and touchscreens. Existing systems operate independently of touch interfaces, require extensive calibration, and do not account for postural context: a virtual keyboard is no more useful when the user's hands are at their sides or when the user is not directing gaze at the input region.

The Unmet Need

There remains a need for a system that (a) operates entirely conventionally under normal conditions, (b) selectively activates auxiliary sensing only when primary-signal confidence is insufficient, (c) does so through a wearable device incapable of persistent recording, display, or surveillance, (d) supports haptic glove input when postural context is appropriate, and (e) degrades gracefully when auxiliary devices are absent or unavailable.

Summary of the Invention

The present invention introduces an opportunistic multimodal input framework comprising:

1. A primary input device comprising a capacitive touchscreen and associated signal-processing pipeline.
2. A secondary wearable constraint device—exemplified as lightweight non-recording glasses—that provides real-time hand-motion and postural context data when activated.
3. An optional tertiary haptic glove input device that encodes finger-motion as symbolic input.
4. A confidence monitor that computes a per-event confidence score from the primary device and activates auxiliary channels when that score falls below a dynamically adjusted threshold.
5. A fusion and constraint evaluation module that integrates signals from active channels to determine whether a candidate input event should be accepted, rejected, or queued pending confirmation.

A central architectural principle is *opportunistic activation*: auxiliary devices impose no overhead during normal operation and are engaged only when needed. The system therefore presents no perceptible latency penalty under the conditions in which a standard touchscreen already performs adequately.

A second architectural principle is *principled incapacity*: the secondary wearable device is defined not only by what it does—provide constraint data—but by what it is architecturally prevented from doing. It stores no imagery, transmits no raw sensor data, provides no display surface, and is incapable of producing a persistent record of the environment. This transforms it from a surveillance instrument into a measuring device, analogous in social character to a computer mouse or stylus.

Detailed Description of Preferred Embodiments

System Overview

Figure 1 illustrates the high-level architecture. The primary input device generates a stream of candidate touch events. A confidence monitor evaluates each event against current environmental and signal-quality parameters. If confidence is sufficient, the event is forwarded directly to the host application. If confidence is insufficient, the constraint channel is activated and the event is held pending cross-modal validation.

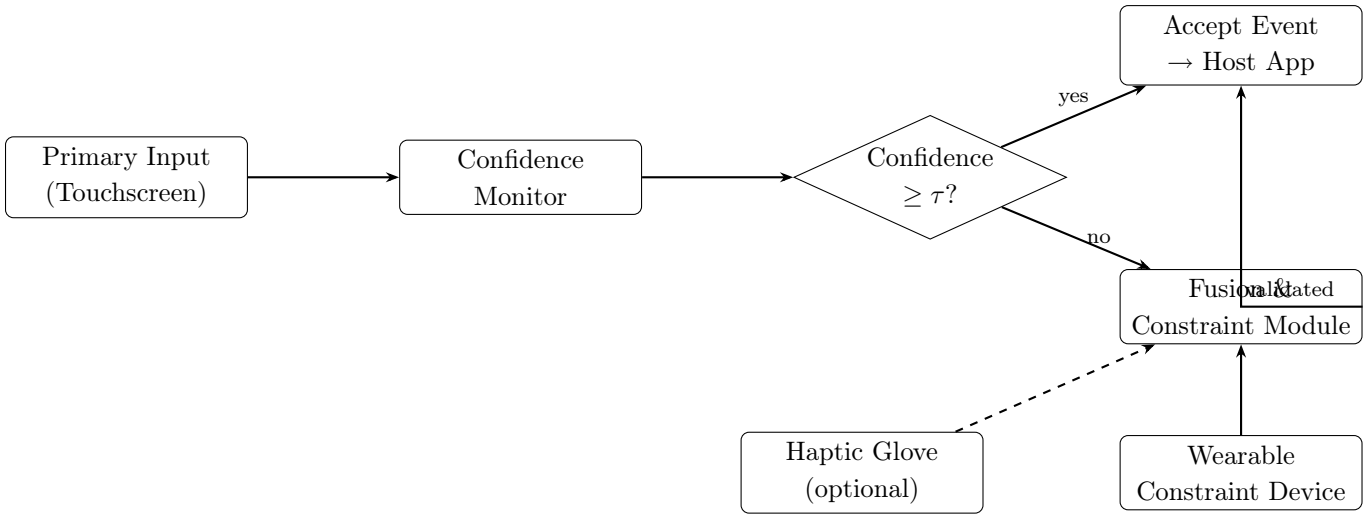

Figure 1: High-level system architecture. Dashed line indicates optional channel. The wearable constraint device is activated only when confidence falls below threshold τ.

Primary Input Device

The primary input device comprises a mutual-capacitance sensing grid, a touch-event detection module, and a spatial-temporal signal-processing unit. The device produces a stream of candidate input events:

$$\mathcal{E} = \{\, e_i = (x_i,\, y_i,\, t_i,\, s_i,\, \sigma_i) \}$$

where (x_i, y_i) denotes screen coordinates, t_i the detection timestamp, s_i the integrated signal strength, and σ_i a per-event confidence score computed from signal morphology, contact-area distribution, approach velocity (where derivable), and historical false-positive rate under current environmental conditions.

The confidence score $\sigma_i \in [0, 1]$ is produced by a local classifier trained on labelled examples of intentional touch, raindrop impact, palm contact, vibrational artefact, and conductive contamination. Under normal operating conditions the vast majority of events will have $\sigma_i \geq \tau$, where τ is the current acceptance threshold, and will proceed directly to the host application without consulting any auxiliary channel.

Confidence Monitor and Threshold Adaptation

The confidence monitor maintains a sliding-window estimate of the ambient false-positive rate $\hat{f}(t)$ and a global environmental state vector $\mathbf{e}(t)$ incorporating readings from the device's onboard sensors: barometric pressure, accelerometer, gyroscope, and optionally a dedicated moisture detector at the screen perimeter.

The acceptance threshold is adapted as:

$$\tau(t) = \tau_0 + \alpha \cdot \hat{f}(t) + \beta \cdot v(t)$$

where τ_0 is a baseline threshold, $v(t)$ is the RMS vibration magnitude from the accelerometer, and α, β are tunable weights. Under calm, dry conditions $\tau(t) \approx \tau_0$ and auxiliary channels are rarely or never invoked. Under rain or vibration, $\tau(t)$ rises, causing more events to be held pending cross-modal confirmation.

The confidence monitor also emits an *activation signal* $A(t) \in \{0, 1\}$ transmitted to the secondary wearable device. When $A(t) = 0$, the wearable enters a low-power idle state. When $A(t) = 1$, the wearable activates its sensing pipeline. This ensures the wearable consumes negligible power during normal operation.

Secondary Wearable Constraint Device

Physical Form Factor. The preferred embodiment of the secondary device takes the form of lightweight glasses. Alternative embodiments include a headband, an ear-mounted device, or a shoulder-mounted sensor. The common requirement across all embodiments is that the device provides a viewpoint geometrically distinct from the primary screen—one from which the user's hands and the screen surface are simultaneously observable—without requiring direct line-of-sight from the screen itself.

Sensing Hardware. The device comprises one or more of the following sensing modalities:

- A monocular or stereo RGB or monochrome optical sensor.
- An infrared illuminator and corresponding IR sensor, enabling operation in low-light environments without visible light emission.
- A time-of-flight (ToF) depth sensor providing per-pixel range estimates.
- An inertial measurement unit (IMU) for head-pose estimation.

Principled Incapacity. The device is architecturally constrained to prevent recording, storage, and display. Specifically:

a. No non-volatile storage is present. Raw sensor data is processed in working memory and discarded within a bounded time window, not to exceed 500 milliseconds.

b. No display surface, projector, or optical combiner is present.

c. The communication interface transmits only derived constraint signals—hand position, velocity, finger-extension estimates, and postural-context flags—not raw imagery or depth maps.

d. Hardware attestation mechanisms, implemented in a secure element, certify to paired devices that no recording capability is present and that all transmitted data conforms to the derived-signal specification.

These constraints are enforced at the hardware level, not merely at the software level, so that the absence of recording is verifiable rather than merely asserted.

Constraint Signal. The wearable produces a constraint signal stream:

$$\mathcal{C} = \left\{ c_j = (\mathbf{p}_j,\, \mathbf{v}_j,\, \phi_j,\, t_j,\, \psi_j) \right\}$$

where $\mathbf{p}_j$ is the estimated 3-D position of the fingertip in a shared coordinate frame, $\mathbf{v}_j$ is its velocity vector, ϕ_j is the finger-extension confidence (0 = fully retracted, 1 = fully extended), t_j

is the timestamp, and ψ_j is a postural-context flag encoding whether the user's hands are in a configuration favourable for deliberate input (see §4.6).

Fusion and Constraint Evaluation Module

Admissibility Function. The fusion module evaluates held candidate events against the constraint stream. An event e_i is admitted if and only if:

$$A(e_i, \mathcal{C}) = 1$$

where the admissibility function A is defined as the conjunction of three conditions:

Temporal correlation. There exists a constraint observation $c_j \in \mathcal{C}$ such that $|t_i - t_j| \leq \Delta t$, where Δt is a configurable temporal window (default: 120 ms).

Spatial intersection. The projected screen-plane position of $\mathbf{p}_j$ lies within a tolerance radius r of (x_i, y_i) in screen coordinates (default: $r = 8\,\text{mm}$).

Motor plausibility. The velocity profile $\mathbf{v}_j$ and finger-extension sequence leading to c_j are consistent with a voluntary reach-and-contact motion, as assessed by a lightweight kinematic classifier. Criteria include: approach velocity within the range $[v_{\min}, v_{\max}]$ characteristic of deliberate finger contact, monotonic decrease in finger-to-screen distance over the preceding window, and absence of lateral scatter inconsistent with directed motion.

Events failing any condition are rejected. Events satisfying all three are promoted to the host application as validated input.

Rejection Taxonomy. Rejected events fall into distinguishable classes:

- *Environmental noise*: events with no temporally proximate constraint observation (e.g., raindrops, condensation).
- *Vibrational artefacts*: events with coincident constraint observations but lacking motor plausibility (e.g., involuntary contact caused by device vibration).
- *Non-directed contact*: events with a present hand but lacking directed approach (e.g., palm resting on screen edge).

This taxonomy is retained in an ephemeral in-memory log for adaptive threshold tuning. The log is not persisted beyond the current interaction session.

Postural Context Assessment

The postural-context flag ψ_j encodes conditions under which haptic-glove input is reliable. The wearable assesses postural context by estimating:

1. *Hand elevation*: whether the user's hands are raised to a position consistent with active typing or gesture input, as opposed to hanging at the sides.
2. *Gaze alignment*: whether head pose, as estimated from the IMU, is directed toward the expected input region. This estimate is coarse and does not require eye tracking.
3. *Stability*: whether hand and head motion are below a vibration threshold indicating a stable input posture.

$\psi_j = 1$ when all three conditions are met; $\psi_j = 0$ otherwise. The haptic glove input channel is enabled only when $\psi_j = 1$, preventing spurious glove-input events from being generated when the user is walking, gesturing conversationally, or otherwise not in an input posture.

Tertiary Haptic Glove Input Channel

Overview. Haptic gloves encode finger flexion, extension, and contact events as symbolic input. They do not require physical contact with a screen surface and thus extend the input modality to environments where screen contact is impractical: standing, outdoor use with bulky gloves, or hands-full scenarios where only one hand is free.

Integration with Constraint Framework. The haptic glove channel participates in the same constraint framework as touch events. Glove-generated events are treated as candidate inputs and are subject to postural-context gating via ψ_j. A virtual keyboard or gesture vocabulary is anchored to the user's physical configuration as assessed by the wearable, so that the mapping between finger motion and intended symbol remains stable as long as $\psi_j = 1$.

When posture degrades ($\psi_j \to 0$), the glove channel is suspended rather than continuing to generate potentially incorrect input. This is the key distinction from prior air-typing systems, which either ignore postural context entirely or require explicit user activation and deactivation.

Motor Adaptation. Over time the fusion module accumulates statistics on the user's characteristic approach angles, velocities, and inter-key timing. A personalised motor model $\mathcal{M}_u$ is maintained in local, encrypted storage on the host device (not the wearable). This model progressively tightens the admissible region for each user's input events, reducing false positives and increasing recognition accuracy for that user's specific motor signature. The model is device-local and is not transmitted.

Operational Modes

Normal Mode. The touchscreen operates conventionally. No wearable communication occurs. The wearable is in low-power idle. All candidate events have $\sigma_i \geq \tau(t)$ and are forwarded directly to the host application. This is the predominant operational mode.

Rain and Moisture Rejection Mode. Activated when the moisture detector or the accelerometer pattern indicates wet conditions, or when $\hat{f}(t)$ exceeds a moisture-onset threshold. In this mode:

- $\tau(t)$ is elevated.
- The wearable is activated.
- Touch events require cross-modal confirmation before being admitted.
- Events with no temporally proximate constraint observation are discarded.

The user experiences transparent operation: the screen continues to respond to deliberate touches while ignoring raindrops, at the cost of a modest ($\leq 120\,\text{ms}$) confirmation latency for events that require cross-modal validation.

Vibration Rejection Mode. Activated when accelerometer RMS exceeds a vibration threshold, as occurs on trains, aircraft, or during earthquakes. In this mode:

- $\tau(t)$ is elevated by the $\beta \cdot v(t)$ term.
- The wearable is activated.
- Motor plausibility filtering is applied with tightened velocity bounds, since intentional touch approaches during vibration have distinctive profiles (the user braces and reaches deliberately) distinguishable from vibration- induced artefacts (which lack directed approach trajectories).

Glove and Indirect Contact Mode. Activated when s_i (signal strength) falls below a glove-detection threshold, indicating a non-conductive or partially conductive contact. In this mode:

- The wearable becomes the primary signal source.
- Contact is inferred from finger trajectory rather than surface perturbation.
- The screen's electrostatic grid continues to provide coarse positional data when available.

Haptic Glove Input Mode. Activated when haptic gloves are paired and $\psi_j = 1$. The screen may be off or absent. Symbolic input is generated from glove-encoded finger motion, validated against the postural-context flag and the user's motor model. When ψ_j drops to 0 (e.g., the user begins walking), this mode is suspended automatically.

Calibration and Coordinate Frame Registration

For spatial-intersection checking, the constraint signal must be expressed in the same coordinate frame as the screen surface. Calibration is performed at pairing time and periodically thereafter:

1. The user performs a brief calibration gesture: touching a sequence of prompted screen targets.
2. The fusion module records simultaneous (x_i, y_i, t_i) tuples from the screen and $(\mathbf{p}_j, t_j)$ tuples from the wearable.
3. A rigid-body transform $T : \mathbb{R}^3 \to \mathbb{R}^2$ is estimated by least-squares minimisation over the calibration pairs.
4. The transform is stored on the host device and updated incrementally as new validated events accumulate, accommodating slow drift due to changes in wearable fit or device position.

Communication Interface

The wearable communicates with the host device via a low-latency wireless protocol (e.g., Bluetooth Low Energy with a custom constraint-data profile, or Ultra-Wideband for sub-centimetre ranging). The protocol transmits only the derived constraint signal $\mathcal{C}$; no raw imagery is transmitted at any time.

End-to-end latency from wearable sensor capture to constraint-signal receipt at the host device is targeted at $\leq 30\,\text{ms}$, leaving adequate margin within the $120\,\text{ms}$ temporal window for reliable correlation.

Prior Art Considerations

Capacitive Touchscreen Filtering

Single-surface filtering and threshold-adjustment approaches (including "wet touch" and "glove modes" found in existing devices) cannot resolve the fundamental underdetermination problem: environmental stimuli and intentional contacts can be identical at the sensing surface. The present invention differs by introducing an independent sensing modality whose observations cannot be mimicked by rain or vibration, transforming the problem from signal classification to cross-modal constraint satisfaction.

Stylus-Based Input

Active styluses constrain input by requiring a specialised tool with identifiable characteristics. The present invention generalises constraint enforcement to natural hand interaction without requiring a dedicated contact instrument, and additionally addresses environmental noise, which stylus approaches do not.

Optical Gesture Recognition

Prior gesture-recognition systems are designed as primary input modalities and depend on continuous scene capture. The present invention uses optical sensing as a secondary constraint channel activated only when needed, and its wearable device is architecturally incapable of producing a persistent scene record.

Smart Glasses and AR Systems

Existing smart-glasses and augmented-reality platforms emphasise display, annotation, and recording, raising privacy concerns and creating social friction. The present invention defines a device class whose distinguishing feature is the principled *absence* of display and recording capability. This is not a software policy but a hardware constraint, verifiable through attestation.

Swype-Style Gesture Disambiguation

The Swype input method resolves underdetermined gesture trajectories by projecting candidate paths into a constrained linguistic space (the user's lexicon), selecting the interpretation most consistent with language statistics. The present invention employs a structurally analogous strategy at a lower level of the input stack: rather than disambiguating which symbol a trajectory encodes, it disambiguates whether a surface event encodes any intentional act at all. The constraint is physical and postural rather than linguistic, but the underlying principle—that raw input is not trusted on its own and must be brought into agreement with an independent constraint source—is shared.

Mobile Sensor Fusion

Existing sensor-fusion techniques in mobile devices combine co-located sensors (accelerometer, gyroscope, proximity, barometric pressure). The present invention extends sensor fusion across physically separate devices, incorporating an external wearable viewpoint, which is a qualitatively distinct extension not achieved by prior intra-device approaches.

Air-Typing and Haptic Glove Systems

Prior haptic and air-typing systems operate independently of touch interfaces and do not incorporate postural-context gating. The present invention integrates haptic input as one channel within a unified constraint framework and explicitly gates glove-mode activation on a postural-context assessment derived from the wearable, preventing spurious input during ambiguous body configurations.

Claims

1. A system for multimodal input disambiguation comprising: a primary input device including a capacitive touchscreen and a signal-processing unit configured to produce a stream of candidate input events with associated confidence scores; a confidence monitor configured to compute a per-event confidence score and to compare said score to a dynamically adjusted acceptance threshold; and a secondary wearable constraint device configured to produce real-time constraint signals encoding hand position, velocity, and motor plausibility, wherein said secondary device is activated only when said confidence monitor determines that the primary-signal confidence is insufficient.

2. The system of claim 1, wherein the secondary wearable constraint device is architecturally incapable of storing persistent imagery or video data, transmitting raw sensor imagery, or providing a visual display surface.

3. The system of claim 2, wherein the secondary wearable constraint device includes a hardware attestation mechanism that certifies to a paired host device that no recording capability is present and that all transmitted data conforms to a derived-signal specification.

4. The system of claim 1, wherein a candidate input event is promoted to the host application if and only if there exists a constraint observation satisfying temporal correlation within a configurable window, spatial intersection within a tolerance radius in screen coordinates, and motor plausibility as assessed by a kinematic classifier.

5. The system of claim 1, wherein the acceptance threshold is adapted as a function of estimated ambient false-positive rate and RMS vibration magnitude, such that the threshold increases under rain or mechanical vibration.

6. The system of claim 1, wherein the secondary wearable constraint device is placed in a low-power idle state when the confidence monitor determines that primary-signal confidence is sufficient, and is activated by an explicit signal from the confidence monitor when the threshold is exceeded.

7. The system of claim 1, further comprising a tertiary haptic glove input device, wherein glove-generated input events are subject to a postural-context gate derived from the secondary wearable constraint device, such that glove-mode input is enabled only when the user's hands, head orientation, and motion magnitude satisfy predefined postural-context criteria.

8. The system of claim 7, wherein the postural-context gate assesses hand elevation, coarse gaze alignment estimated from an inertial measurement unit, and hand-motion stability, and wherein glove-mode input is automatically suspended when any postural-context criterion is not met.

9. The system of claim 1, wherein a coordinate-frame registration transform is estimated from simultaneous paired observations of calibration touch events and corresponding wearable constraint observations, and is updated incrementally as validated events accumulate during normal operation.

10. The system of claim 1, wherein rejected candidate events are classified into an ephemeral in-memory rejection taxonomy comprising environmental noise, vibrational artefacts, and non-directed contact, and wherein said taxonomy informs adaptive threshold adjustment without being persisted beyond the current interaction session.

11. The system of claim 1, wherein a personalised motor model is maintained in local encrypted storage on the host device and progressively refines the admissible kinematic region for individual user input events, improving recognition accuracy over time without transmitting user data to external systems.

12. A method for opportunistic multimodal input disambiguation comprising: generating candidate input events from a primary capacitive touchscreen; computing a confidence score for each event; comparing each confidence score to an adaptive threshold; forwarding events whose confidence meets or exceeds the threshold directly to the host application; activating a secondary non-recording wearable sensor upon detecting that confidence is insufficient; receiving constraint signals from the secondary sensor encoding hand position, velocity, and motor plausibility; and admitting or rejecting held candidate events based on cross-modal agreement between screen signals and constraint signals.

13. The method of claim 12, wherein activating the secondary sensor comprises transmitting an explicit activation signal to a wearable device that was previously in a low-power idle state, such that the wearable consumes negligible power during normal operation.

14. The method of claim 12, further comprising assessing postural context from the secondary sensor and enabling a haptic glove input channel only when the assessed postural context satisfies predefined criteria.

15. A wearable constraint device comprising: an optical or infrared sensor; an on-device processing unit configured to estimate hand position, velocity, and finger-extension state from transient

sensor data; a communication interface configured to transmit derived constraint signals to a paired host device; and hardware limitations that prevent persistent storage of sensor imagery, raw data transmission, and visual display output, wherein the device is designed to serve as an ephemeral constraint provider for touch-input disambiguation rather than as a primary input, display, or recording device.

Brief Description of the Drawings

Figure 1 is a block diagram illustrating the high-level architecture of the system, including the primary input device, confidence monitor, fusion module, wearable constraint device, and optional haptic glove channel.

Additional figures, including a timing diagram for opportunistic activation, a spatial-intersection schematic for coordinate-frame registration, and a postural- context state diagram, are contemplated for a non-provisional filing.

Appendix: Theoretical Framing

The invention can be situated within a broader theory of *constraint-enforced input systems*. In this framing, a touchscreen operating alone attempts to infer intentional input from a single recurrence channel: repeated perturbations of an electrostatic field. The problem is that the field cannot distinguish the sources of perturbation. The invention resolves this by coupling the primary recurrence channel to a secondary constraint channel, so that an input event is admitted only when it satisfies the structure imposed by both.

This is structurally analogous to Swype-style keyboard input, in which a raw gesture trajectory is disambiguated by projecting it into a constrained linguistic space. The present invention performs an analogous disambiguation at the level of physical plausibility rather than symbolic identity: instead of asking *which word does this trajectory encode*, it asks *does this surface event encode any deliberate act at all*. The constraint is physical and kinematic rather than lexical, but the structural principle is the same: raw signals are not trusted in isolation; they must agree with an independent constraint before being admitted.

The device itself embodies a corresponding design principle. Smart glasses fail partly because they are underspecified: users and bystanders cannot determine what they are doing. The wearable device described herein is overspecified in the opposite direction—it is defined as much by what it cannot do as by what it can. A device whose architectural incapacities are verifiable through hardware attestation occupies a different social category from a general-purpose recording device. It is legible in the way that a keyboard, mouse, or stylus is legible: its function is narrow, its

operation is transient, and it produces no durable record of the world.

This document constitutes a draft patent specification prepared for conceptual documentation purposes under the Canadian Patent Act, R.S.C. 1985, c. P-4. It has not been filed with the Canadian Intellectual Property Office (CIPO) and confers no legal priority date. A formal application may be filed with CIPO at any time prior to public disclosure to establish priority. All technical specifications are subject to revision. The inventor is resident in Canada.